

## Adding 3 and 4 Force Vectors

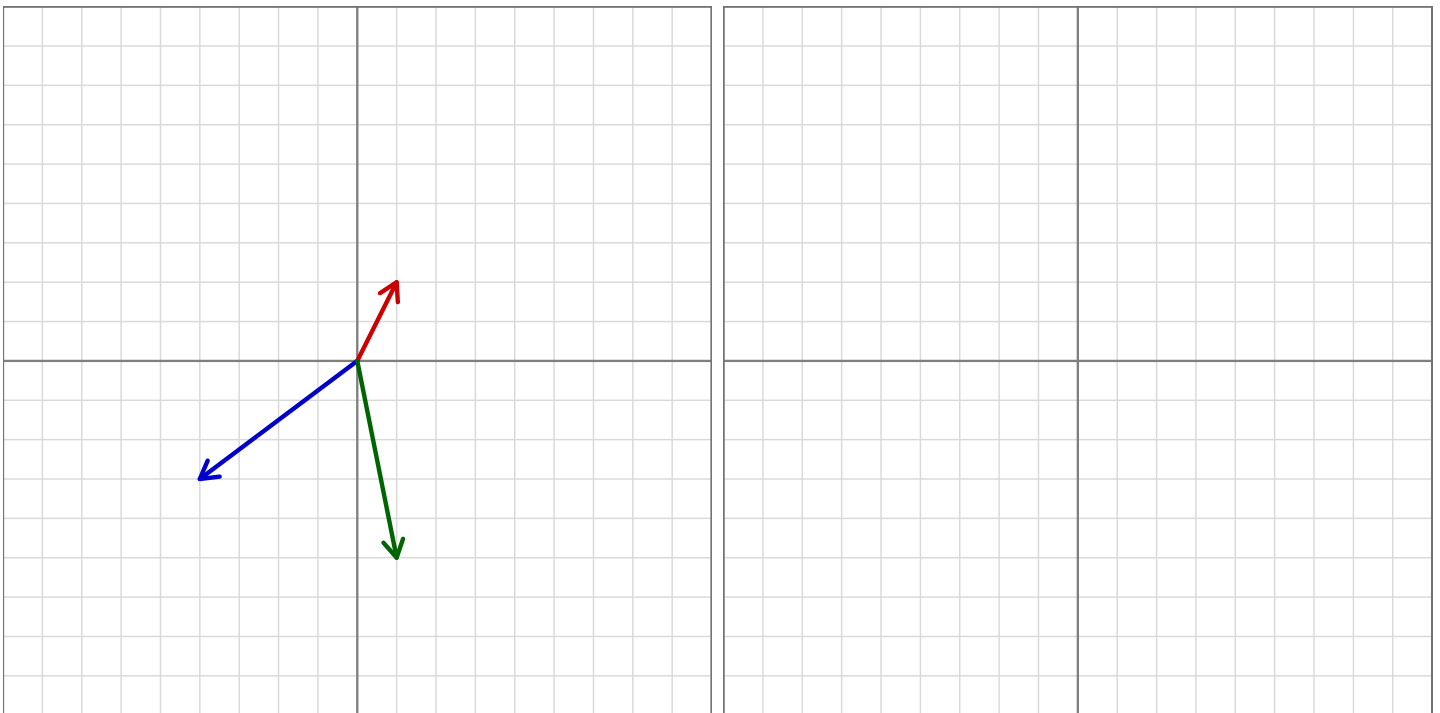
For each problem, use the left grid to read the given vectors. On the right grid, redraw the vectors **tip-to-tail** to make a chain. Draw the resultant from the beginning of the chain to its end.

### Part A: Three Force Vectors

#### Problem 1

$$\mathbf{F1} = \langle -4, -3 \rangle \quad \mathbf{F2} = \langle 1, 2 \rangle \quad \mathbf{F3} = \langle 1, -5 \rangle$$

On the right grid, draw the chain and the resultant vector.



Resultant vector:  $\langle \underline{\hspace{1cm}}, \underline{\hspace{1cm}} \rangle$

Assuming these vectors represent force in newtons, what is the magnitude of the net (resultant) force?

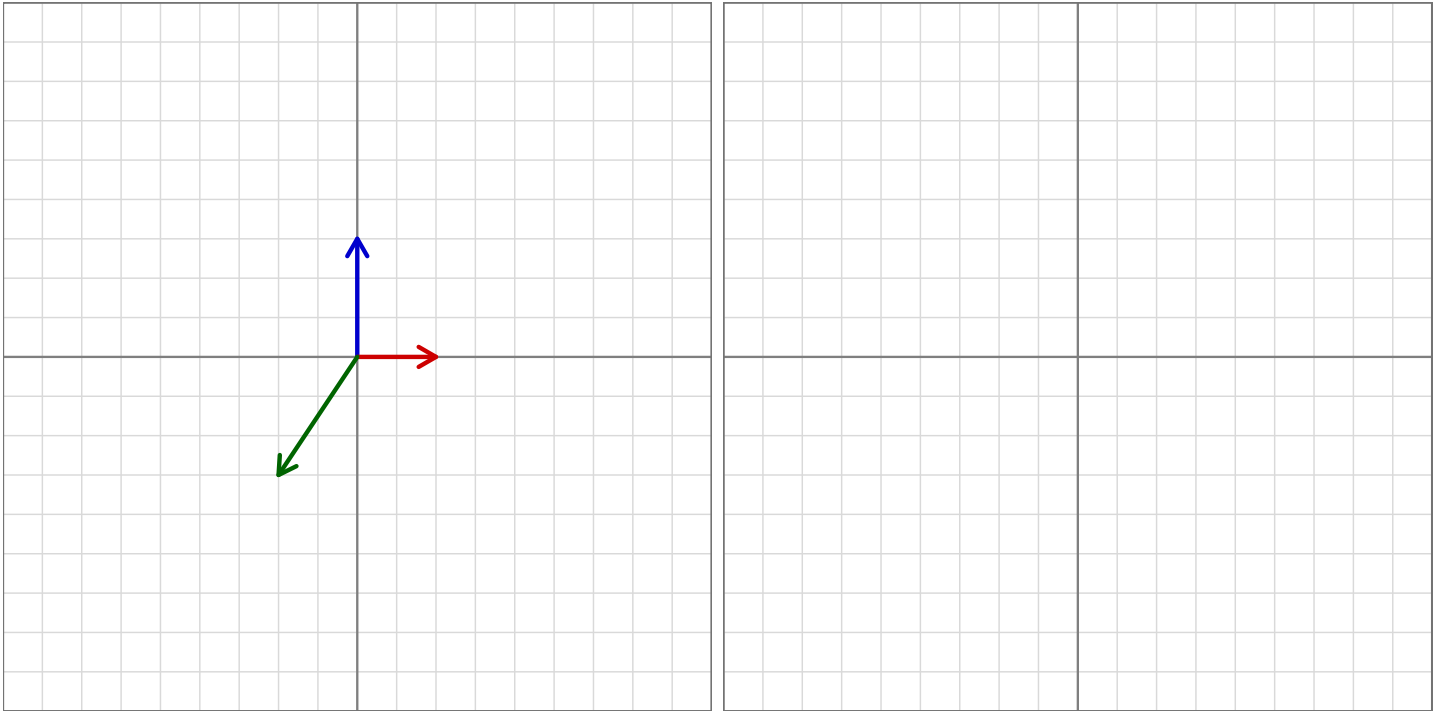
Is  $\sum F = 0$ ?

Are these forces balanced or unbalanced? Balanced / Unbalanced

## Problem 2

$$\mathbf{F1} = \langle 0, 3 \rangle \quad \mathbf{F2} = \langle 2, 0 \rangle \quad \mathbf{F3} = \langle -2, -3 \rangle$$

On the right grid, draw the chain and the resultant vector.



Resultant vector:  $\langle \underline{\hspace{1cm}}, \underline{\hspace{1cm}} \rangle$

Assuming these vectors represent force in newtons, what is the magnitude of the net (resultant) force?

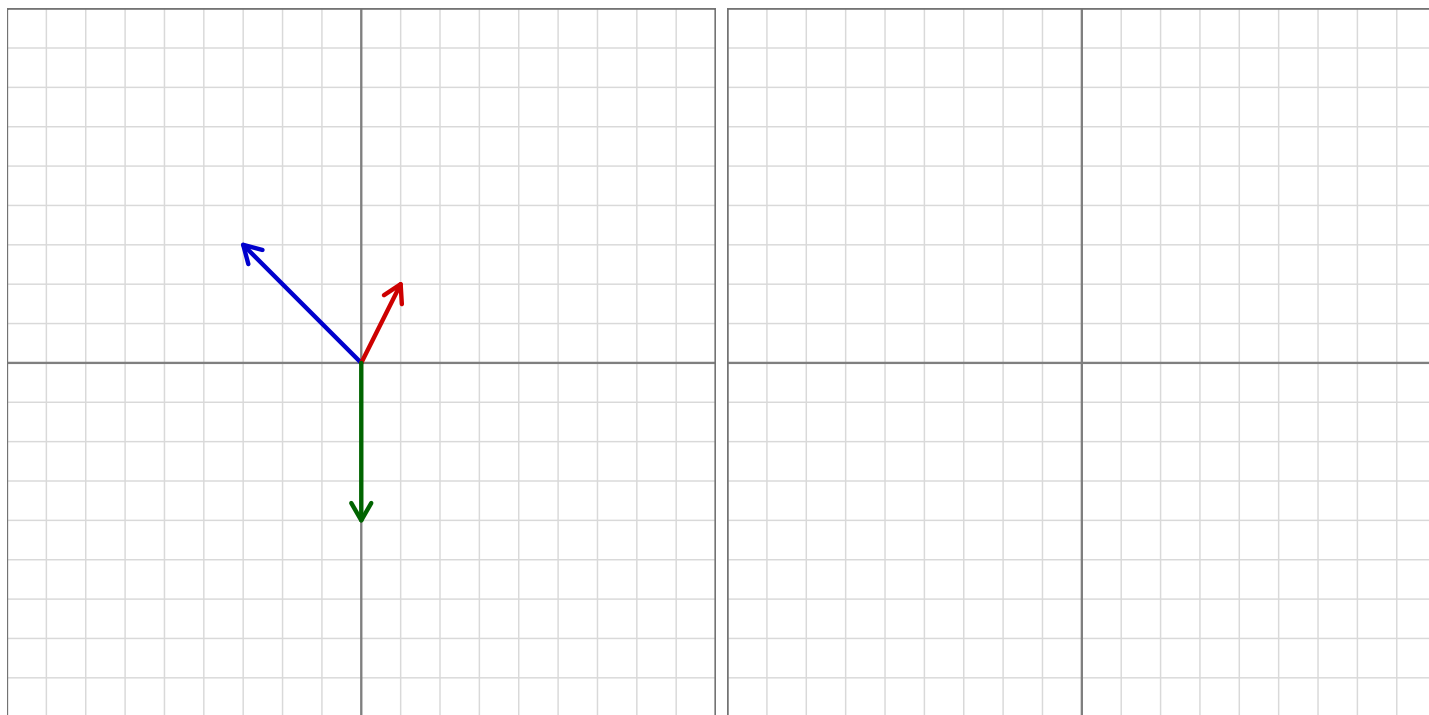
Is  $\sum F = 0$ ?

Are these forces balanced or unbalanced? Balanced / Unbalanced

### Problem 3

$$\mathbf{F1} = \langle -3, 3 \rangle \quad \mathbf{F2} = \langle 1, 2 \rangle \quad \mathbf{F3} = \langle 0, -4 \rangle$$

On the right grid, draw the chain and the resultant vector.



Resultant vector:  $\langle \underline{\hspace{1cm}}, \underline{\hspace{1cm}} \rangle$

Assuming these vectors represent force in newtons, what is the magnitude of the net (resultant) force?

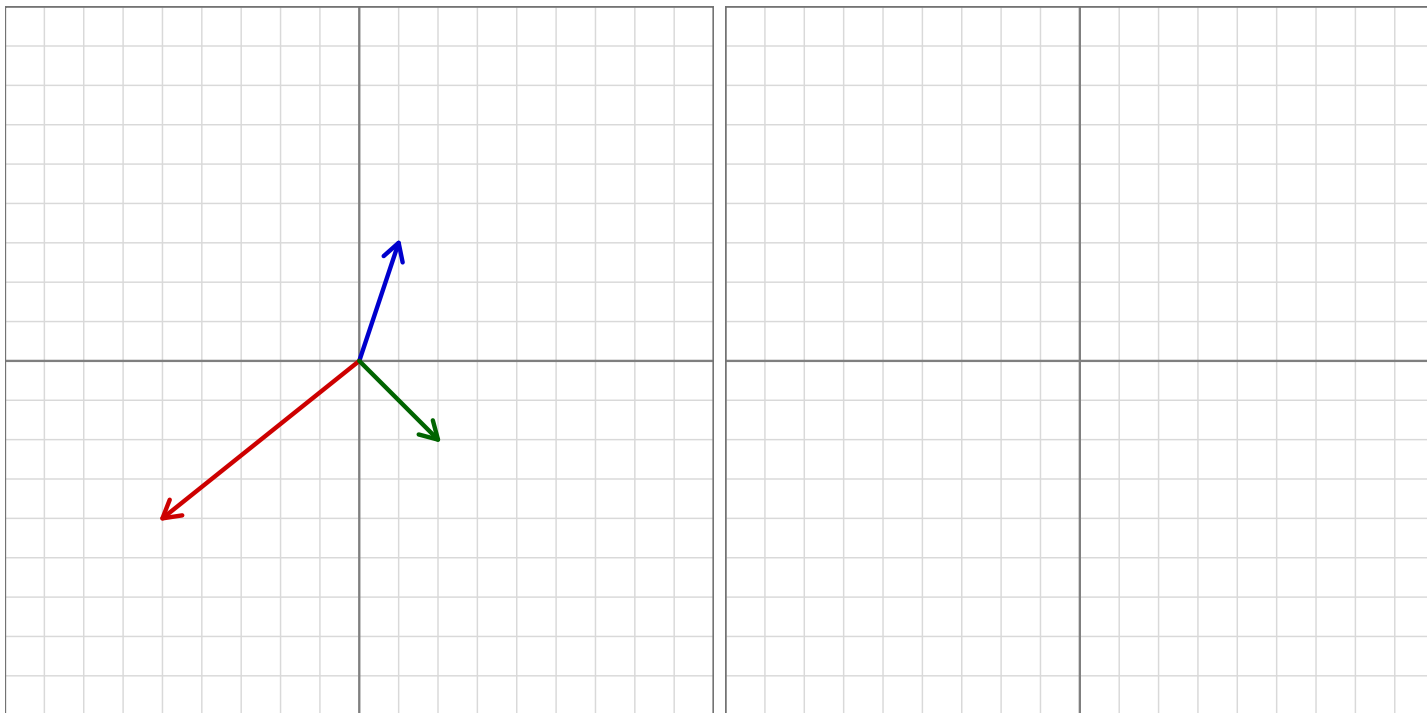
Is  $\sum F = 0$ ?

Are these forces balanced or unbalanced? Balanced / Unbalanced

#### Problem 4

$$\mathbf{F1} = \langle 1, 3 \rangle \quad \mathbf{F2} = \langle -5, -4 \rangle \quad \mathbf{F3} = \langle 2, -2 \rangle$$

On the right grid, draw the chain and the resultant vector.



Resultant vector:  $\langle \underline{\hspace{1cm}}, \underline{\hspace{1cm}} \rangle$

Assuming these vectors represent force in newtons, what is the magnitude of the net (resultant) force?

Is  $\sum F = 0$ ?

Are these forces balanced or unbalanced? Balanced / Unbalanced

### Problem 5

$$\mathbf{F}_1 = \langle -1, -4 \rangle \quad \mathbf{F}_2 = \langle -1, 0 \rangle \quad \mathbf{F}_3 = \langle 2, 4 \rangle$$

On the right grid, draw the chain and the resultant vector.



Resultant vector:  $\langle \underline{\hspace{1cm}}, \underline{\hspace{1cm}} \rangle$

Assuming these vectors represent force in newtons, what is the magnitude of the net (resultant) force?

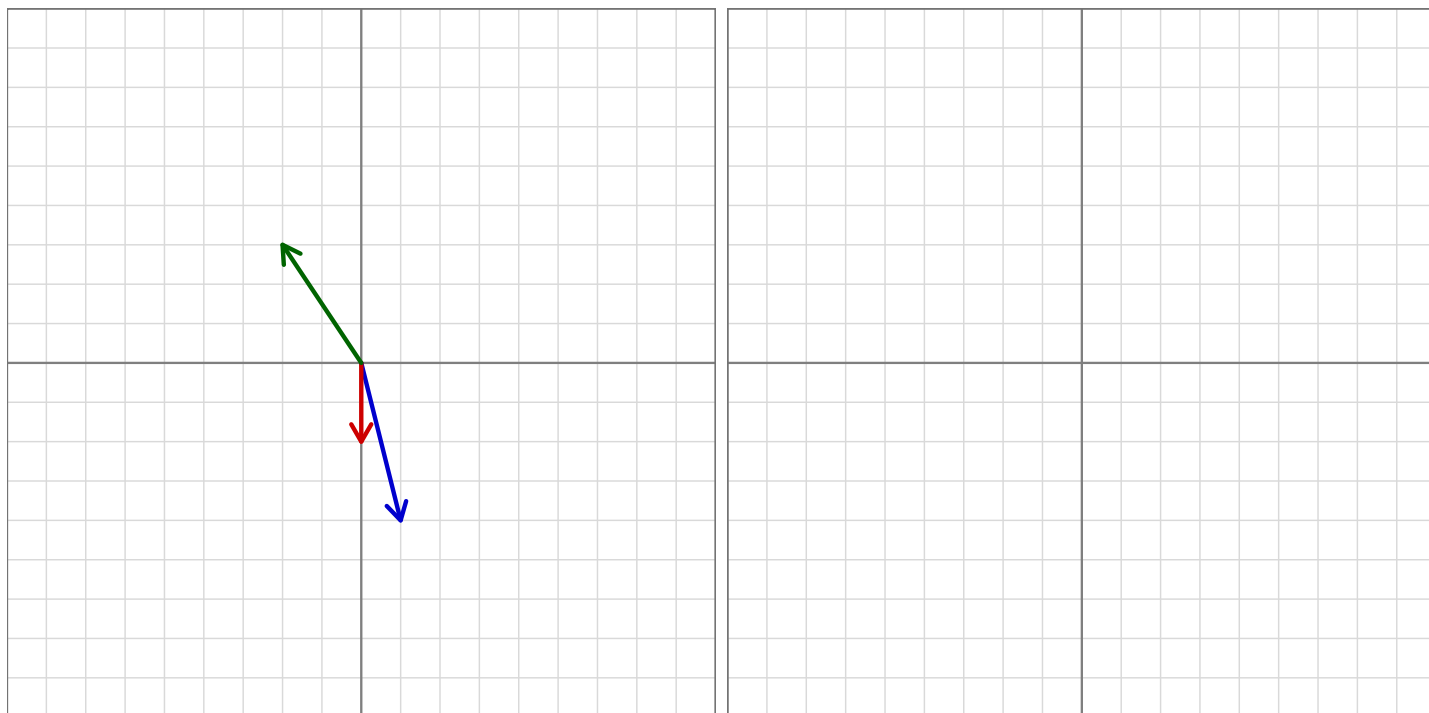
Is  $\sum F = 0$ ?

Are these forces balanced or unbalanced? Balanced / Unbalanced

### Problem 6

$$\mathbf{F1} = \langle 1, -4 \rangle \quad \mathbf{F2} = \langle 0, -2 \rangle \quad \mathbf{F3} = \langle -2, 3 \rangle$$

On the right grid, draw the chain and the resultant vector.



Resultant vector:  $\langle \underline{\hspace{1cm}}, \underline{\hspace{1cm}} \rangle$

Assuming these vectors represent force in newtons, what is the magnitude of the net (resultant) force?

Is  $\sum F = 0$ ?

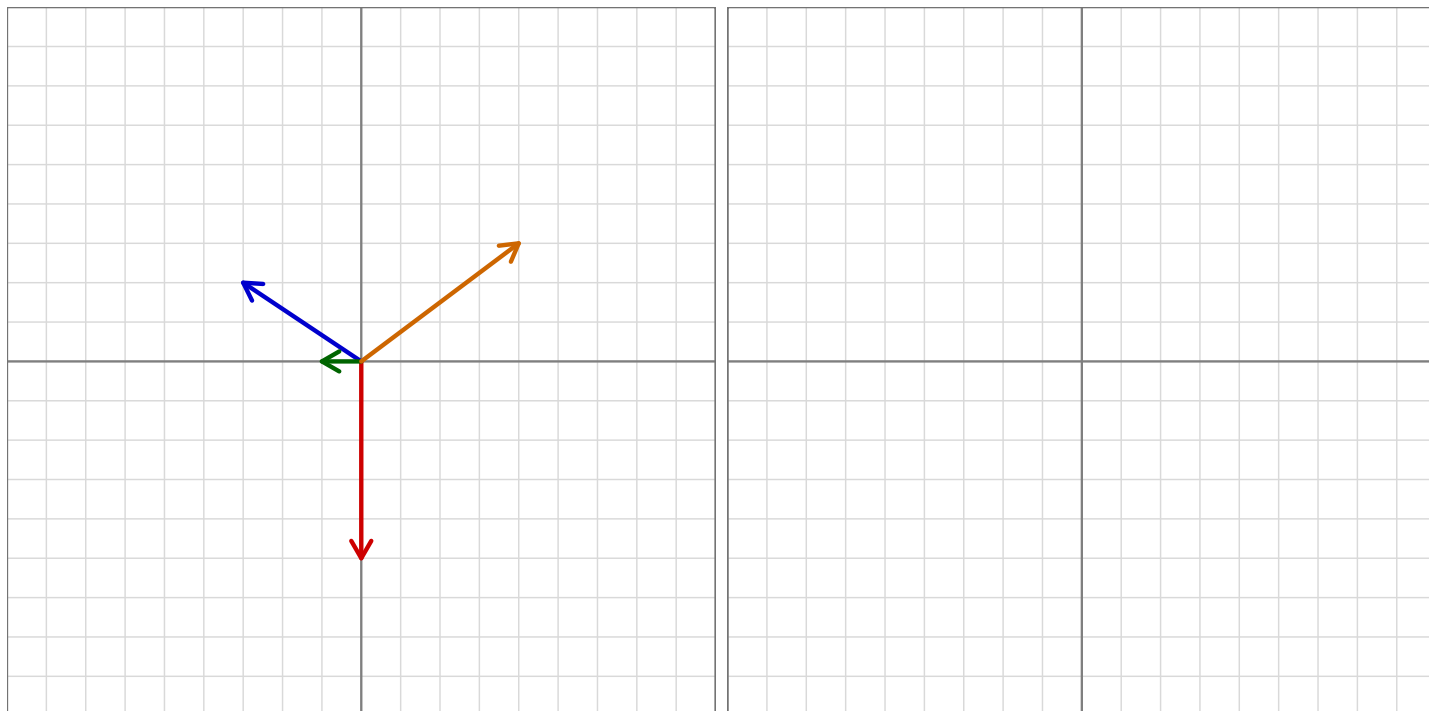
Are these forces balanced or unbalanced? Balanced / Unbalanced

## Part B: Four Force Vectors

### Problem 7

$$\mathbf{F}_1 = \langle -3, 2 \rangle \quad \mathbf{F}_2 = \langle 0, -5 \rangle \quad \mathbf{F}_3 = \langle -1, 0 \rangle \quad \mathbf{F}_4 = \langle 4, 3 \rangle$$

On the right grid, draw the chain and the resultant vector.



Resultant vector:  $\langle \underline{\hspace{1cm}}, \underline{\hspace{1cm}} \rangle$

Assuming these vectors represent force in newtons, what is the magnitude of the net (resultant) force?

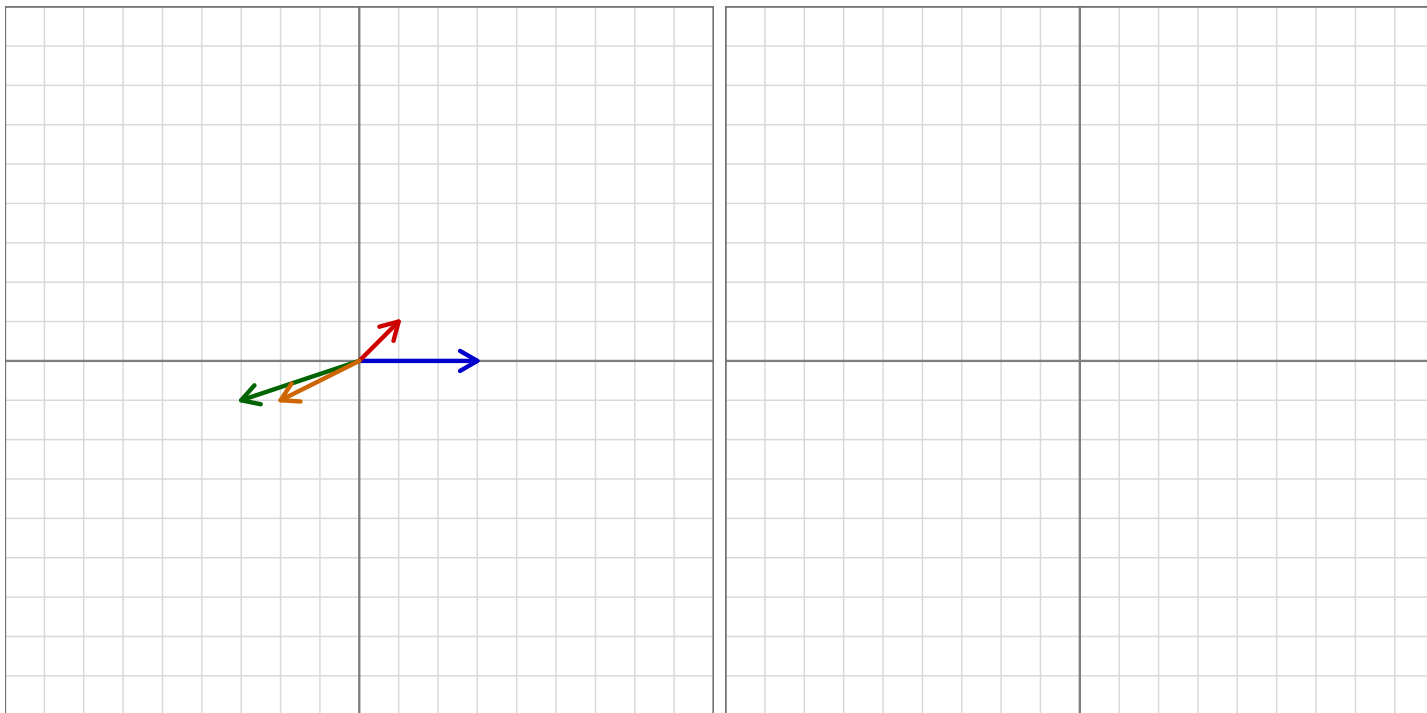
Is  $\sum F = 0$ ?

Are these forces balanced or unbalanced? Balanced / Unbalanced

### Problem 8

$$\mathbf{F1} = \langle 3, 0 \rangle \quad \mathbf{F2} = \langle 1, 1 \rangle \quad \mathbf{F3} = \langle -3, -1 \rangle \quad \mathbf{F4} = \langle -2, -1 \rangle$$

On the right grid, draw the chain and the resultant vector.



Resultant vector:  $\langle \underline{\hspace{1cm}}, \underline{\hspace{1cm}} \rangle$

Assuming these vectors represent force in newtons, what is the magnitude of the net (resultant) force?

Is  $\sum F = 0$ ?

Are these forces balanced or unbalanced? Balanced / Unbalanced

### Problem 9

$$\mathbf{F}_1 = \langle -2, 5 \rangle \quad \mathbf{F}_2 = \langle 3, -5 \rangle \quad \mathbf{F}_3 = \langle 1, -3 \rangle \quad \mathbf{F}_4 = \langle -1, 2 \rangle$$

On the right grid, draw the chain and the resultant vector.



Resultant vector:  $\langle \underline{\hspace{1cm}}, \underline{\hspace{1cm}} \rangle$

Assuming these vectors represent force in newtons, what is the magnitude of the net (resultant) force?

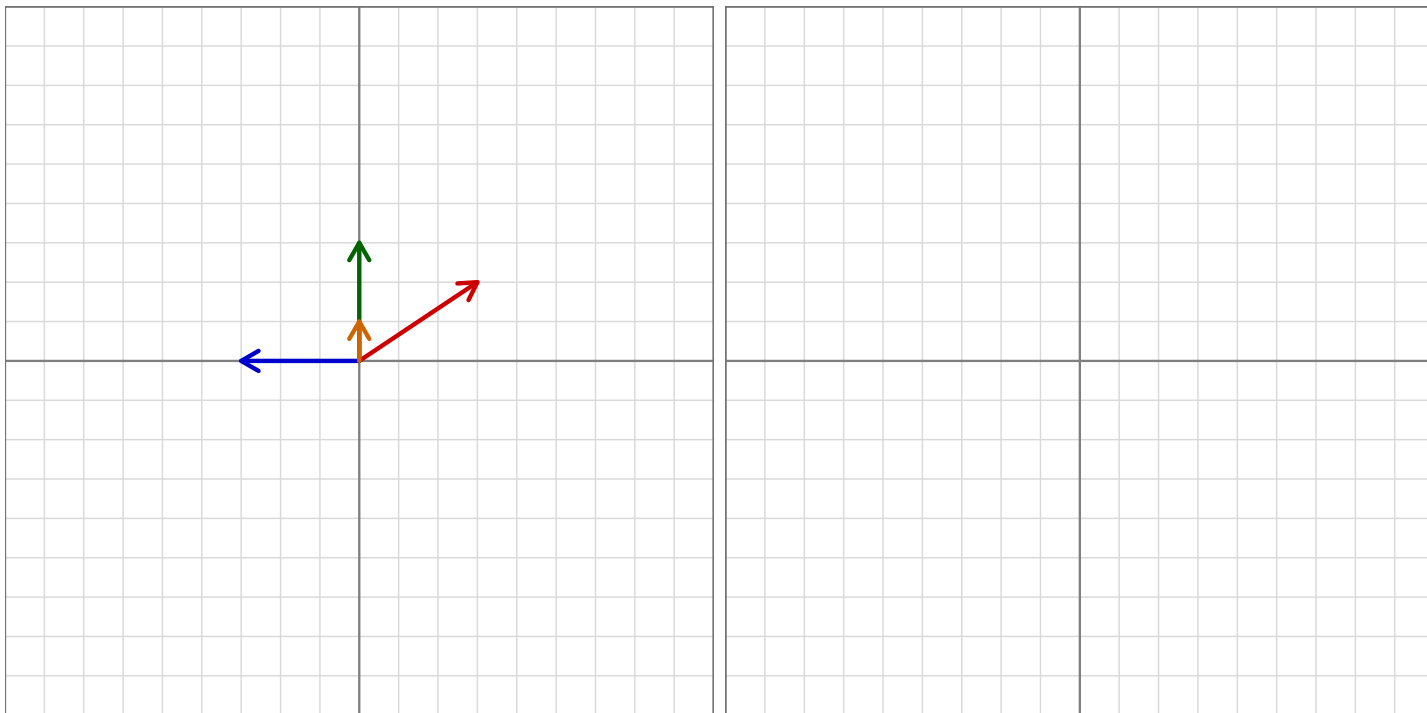
Is  $\sum F = 0$ ?

Are these forces balanced or unbalanced? Balanced / Unbalanced

### Problem 10

$$\mathbf{F}_1 = \langle -3, 0 \rangle \quad \mathbf{F}_2 = \langle 3, 2 \rangle \quad \mathbf{F}_3 = \langle 0, 3 \rangle \quad \mathbf{F}_4 = \langle 0, 1 \rangle$$

On the right grid, draw the chain and the resultant vector.



Resultant vector:  $\langle \underline{\hspace{1cm}}, \underline{\hspace{1cm}} \rangle$

Assuming these vectors represent force in newtons, what is the magnitude of the net (resultant) force?

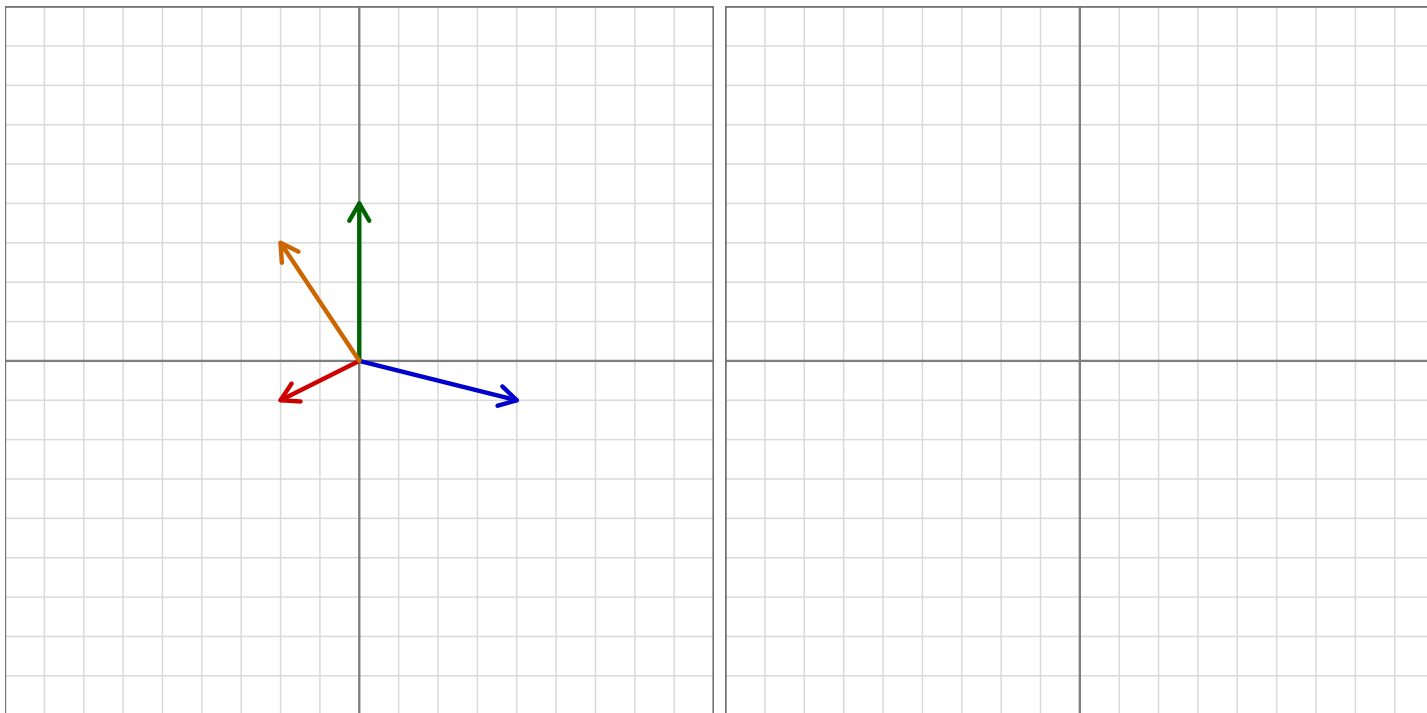
Is  $\sum F = 0$ ?

Are these forces balanced or unbalanced? Balanced / Unbalanced

### Problem 11

$$\mathbf{F}_1 = \langle 4, -1 \rangle \quad \mathbf{F}_2 = \langle -2, -1 \rangle \quad \mathbf{F}_3 = \langle 0, 4 \rangle \quad \mathbf{F}_4 = \langle -2, 3 \rangle$$

On the right grid, draw the chain and the resultant vector.



Resultant vector:  $\langle \underline{\hspace{1cm}}, \underline{\hspace{1cm}} \rangle$

Assuming these vectors represent force in newtons, what is the magnitude of the net (resultant) force?

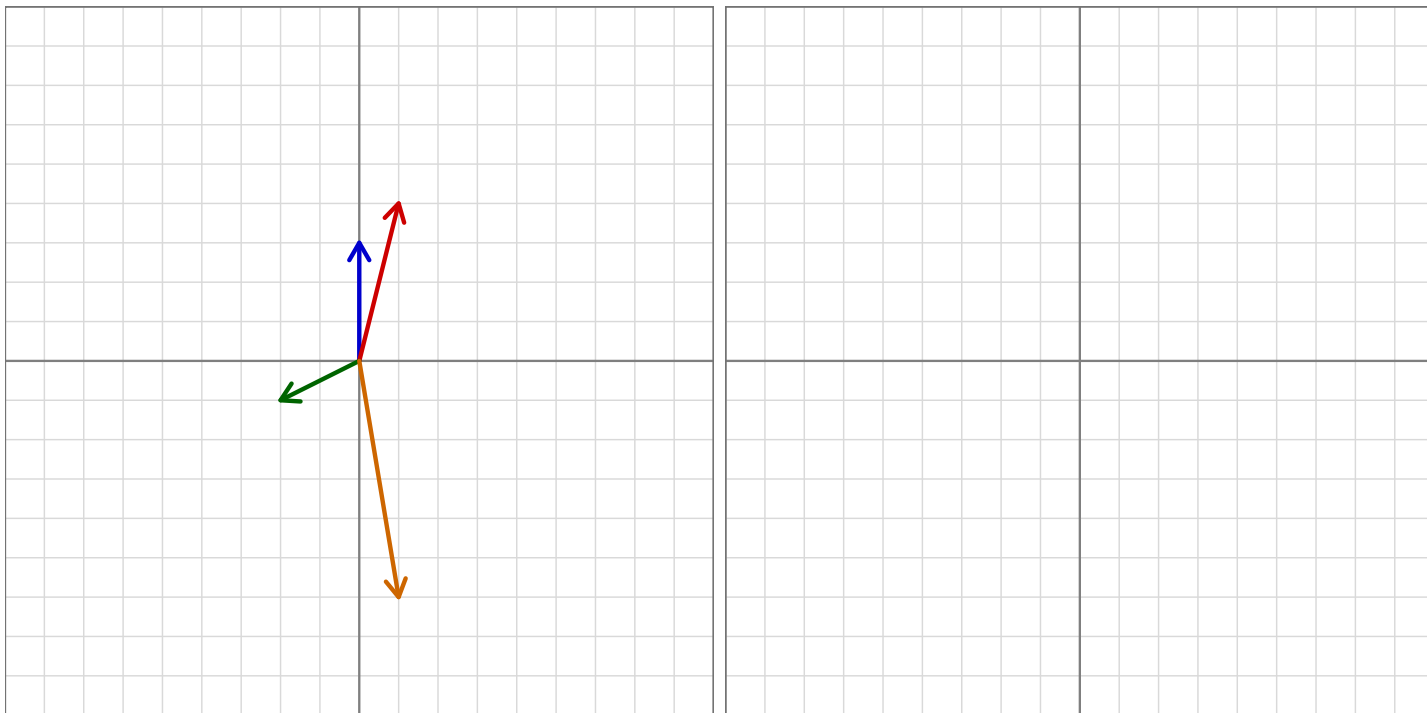
Is  $\sum F = 0$ ?

Are these forces balanced or unbalanced? Balanced / Unbalanced

### Problem 12

$$\mathbf{F1} = \langle 0, 3 \rangle \quad \mathbf{F2} = \langle 1, 4 \rangle \quad \mathbf{F3} = \langle -2, -1 \rangle \quad \mathbf{F4} = \langle 1, -6 \rangle$$

On the right grid, draw the chain and the resultant vector.



Resultant vector:  $\langle \underline{\hspace{1cm}}, \underline{\hspace{1cm}} \rangle$

Assuming these vectors represent force in newtons, what is the magnitude of the net (resultant) force?

Is  $\sum F = 0$ ?

Are these forces balanced or unbalanced? Balanced / Unbalanced